

Cosine-to-WSD Loss-Curve Prediction with an Interpretable MPL-LD Residual Response

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<https://github.com/wjjpku/DL-final>

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Abstract

This report studies a narrow but practically useful loss-curve prediction problem: given a strong Multi-Power Law (MPL) baseline and only a cosine calibration curve, can we improve prediction on warmup-stable-decay (WSD) schedules without fitting any WSD target loss? The main difficulty is that the cosine residual is not a clean schedule-response signal. It also contains smooth errors caused by the LR-dependent parameters of MPL, and a direct one-dimensional projection of the residual onto an LR-drop response feature fails badly. We propose a low-capacity correction,

$$\widehat{L}_s(t) = L_{\text{MPL},s}(t) + a_s \widehat{\kappa}_s \phi_{\lambda_s,s}(t),$$

where the response feature is a causal LR-drop convolution, the response rate is determined by a schedule-only Herfindahl drop concentration q_s , and the locality factor a_s is an explicit support-projection energy fraction rather than a learned gate. The only quantity fitted from loss residuals is one nonnegative scalar $\widehat{\kappa}_s$, estimated after projecting both the cosine residual and the response feature away from the tangent space of MPL’s LR-dependent parameters. On the current public loss-curve repository, the resulting q_2 half-life MPL-LD response improves same-scale WSD-family MAE by -29.88% on average with a -4.67% worst row and 15/15 wins; cross-scale transfer gives -24.95% mean improvement with 30/30 wins. Constant and short-cosine controls remain non-harmful. The evidence supports the correction as a development-stage, mechanism-motivated residual estimator, not as a universal training law.

1 Introduction

Learning-rate (LR) schedules strongly affect the full pretraining loss curve. For schedule search, it is not enough to predict the final loss after training a target schedule: the useful task is to predict the target curve before spending that compute. This report focuses on the following setting:

Use a strong MPL baseline and a source cosine curve to predict WSD-family target loss curves. Target WSD losses are used only for evaluation.

The setting is intentionally restrictive. MPL already models a large fraction of the schedule dependence, so any useful correction must explain a residual that is small, structured, and easy to overfit. Moreover, cosine is a smooth schedule. A residual pattern learned on cosine can mix together two different effects: a transferable LR-drop response and a non-transferable smooth drift caused by MPL parameter error. This is why a high-capacity residual basis, a family-specific route, or a learned gate can look attractive numerically but is not a satisfactory explanation.

The current contribution is a compact residual model with three design constraints.

- The shape of the correction is computed only from the target LR schedule.
- The amplitude is fitted only from the source cosine residual.
- The nuisance removal has an MPL-specific interpretation: it removes local errors in MPL’s LR-dependent tangent directions, rather than an arbitrary low-frequency basis.

The result is a single-formula correction,

$$\widehat{L}_s(t) = L_{\text{MPL},s}(t) + a_s \widehat{\kappa}_s \phi_{\lambda_s,s}(t). \quad (1)$$

Here $L_{\text{MPL},s}$ is the frozen MPL prediction; $\phi_{\lambda_s,s}$ is a causal LR-drop response; λ_s is chosen from the effective concentration of the target LR drops; a_s removes diffuse full-horizon schedule modes; and $\widehat{\kappa}_s$ is the only residual-fitted scalar.

Main findings. The current formula gives the following picture on the repository used in this project. First, raw cosine residual fitting is unsafe: without MPL-LD nuisance removal, the WSD-family MAE worsens by +602.17% on average and wins 0/15 rows. Second, MPL-LD tangent projection recovers a stable transferable signal. Third, replacing the older max-drop response rule by the Herfindahl concentration q_2 and interpreting the locality factor as a support projection gives essentially the same numerical performance, but with a cleaner story and no extra fitted parameters. The current version improves same-scale WSD-family MAE by -29.88% and cross-scale WSD-family MAE by -24.95% , with all WSD rows improving and all controls non-harmful.

2 Problem Setup

Baseline. We use the published MPL form as a frozen baseline. In simplified notation,

$$L_{\text{MPL}}(t) = L_0 + AS(t)^{-\alpha} + B \sum_{k \leq t} \Delta \eta_k G(\eta_k^{-\gamma}(S(t) - S(k))), \quad S(t) = \sum_{i \leq t} \eta_i. \quad (2)$$

The LR-dependent MPL parameters are (B, C, β, γ) , where G is the saturating kernel used by MPL. We do not refit MPL in this report; it is treated as the base predictor whose residual we try to explain.

Data and evaluation. The source calibration curve is `cosine_72000.csv`. The WSD-family targets are five curves at each of three scales 25M, 100M, and 400M: `wsd_20000_24000`, `wsdld_20000_24000`, and three `wsdcon` endpoints. Controls are `cosine_24000`, `constant_24000`, and `constant_72000`. Same-scale evaluation gives $3 \times 5 = 15$ WSD rows; cross-scale evaluation gives $3 \times 2 \times 5 = 30$ WSD rows. We report percentage MAE change relative to MPL:

$$\Delta_{\text{MAE}} = 100 \cdot \frac{\text{MAE}(\widehat{L}) - \text{MAE}(L_{\text{MPL}})}{\text{MAE}(L_{\text{MPL}})}. \quad (3)$$

Negative values are improvements.

Calibration rule. The residual coefficient is fitted on the suffix $t \geq 8000$ of the source cosine curve. This gives $N_{\text{cal}} = 500$ logged calibration points, so the ridge in the one-dimensional projection is $1/N = 0.002$. The target losses are not used to fit λ_s , a_s , $\widehat{\kappa}_s$, or any other quantity.

3 Method

3.1 Causal LR-Drop Response

Let the positive LR drop be

$$d_t = [\eta_{t-1} - \eta_t]_+. \quad (4)$$

The response feature is a causal one-pole recursion in cumulative-LR time:

$$z_t = \exp(-\lambda_s \eta_t) z_{t-1} + d_t, \quad \phi_{\lambda_s,s}(t) = \frac{z_t}{\eta_{\max}}, \quad z_0 = 0. \quad (5)$$

Equivalently,

$$\phi_{\lambda_s, s}(t) = \frac{1}{\eta_{\max}} \sum_{u \leq t} d_u \exp\left(-\lambda_s \sum_{v=u+1}^t \eta_v\right). \quad (6)$$

This feature has three important properties. It is causal, because only past LR drops contribute. It is drop-only, because constant LR intervals do not inject new response. It has a single response channel, so it does not use schedule-family labels or hand-routed cases.

3.2 q_2 Half-Life Response Rate

The response rate λ_s should be schedule-dependent, but not target-loss-dependent. We first convert the positive LR drops into a probability distribution:

$$D_s = \sum_t d_t, \quad p_t = \frac{d_t}{D_s}, \quad q_s = \sum_t p_t^2 = \frac{1}{n_{\text{eff}}}. \quad (7)$$

When the LR drop happens in one step, $q_s = 1$. When the same total drop is spread uniformly over n steps, $q_s = 1/n$. Thus q_s is an effective drop concentration, not a learned label.

The loss curves are logged mainly every $\Delta_{\text{obs}} = 128$ training steps. We therefore anchor the response rate to the observable half-life scale

$$\lambda_{\text{obs}} = \frac{\log 2}{\eta_{\max} \Delta_{\text{obs}}}. \quad (8)$$

The current rule linearly interpolates the response half-life between two observation intervals for diffuse drops and one observation interval for a single sharp drop:

$$H_s = (2 - q_s) \Delta_{\text{obs}}, \quad \lambda_s = \frac{\log 2}{\eta_{\max} H_s} = \frac{\lambda_{\text{obs}}}{2 - q_s}. \quad (9)$$

This is deliberately modest. It does not claim a universal physical relaxation constant; it only says that a sharper LR drop should be resolved on a shorter observed response time than a diffuse LR decay. In the current data, the rule gives $\lambda_s \approx 9.03$ for WSD sharp/linear and $\lambda_s = 18.05$ for the single-step WSD-con targets.

3.3 Support-Projection Locality

Without a locality boundary, a local WSD cooldown correction can be transferred onto a full-horizon diffuse cosine decay. The current formula uses

$$a_s = \mathbf{1}\{D_s > 0\} \frac{\|(I - P_u)m_s\|_2^2}{\|m_s\|_2^2} = \mathbf{1}\{D_s > 0\} \left[1 - \frac{\ell_s}{T_s - W}\right]_+. \quad (10)$$

Here $T_s - W$ is the post-warmup horizon, ℓ_s is the span of the positive-drop support, m_s is the uniform density on that support, and u is the uniform diffuse mode on the full post-warmup horizon. The equality follows from $\|m_s\|_2^2 = 1/\ell_s$ and $\|P_u m_s\|_2^2 = 1/(T_s - W)$. Therefore a_s is not a learned gate: it is the remaining local forcing energy after removing the full-horizon diffuse mode. It is 0.8168 for finite WSD sharp/linear cooldowns, approximately 1 for WSD-con single drops, and 0 for the short-cosine and constant controls.

3.4 MPL-LD Tangent Nuisance Projection

The central identification issue is that the source cosine residual is contaminated by MPL parameter error. Write

$$r_{\text{cos}}(t) = L_{\text{cos}}(t) - L_{\text{MPL,cos}}(t). \quad (11)$$

If the LR-dependent MPL parameters are locally misestimated, then a first-order residual component has the form

$$J_{\text{LD}}(t) \Delta \theta_{\text{LD}}, \quad J_{\text{LD}}(t) = \left[\frac{\partial L_{\text{MPL}}}{\partial \log B}, \frac{\partial L_{\text{MPL}}}{\partial \log C}, \frac{\partial L_{\text{MPL}}}{\partial \log \beta}, \frac{\partial L_{\text{MPL}}}{\partial \log \gamma} \right]. \quad (12)$$

We compute this tangent basis by finite differences of the frozen MPL formula on the source cosine suffix. Let P_{LD} be the orthogonal projection onto the columns of J_{LD} . For a target schedule s , define

$$x_s = (I - P_{\text{LD}})\phi_{\lambda_s, \text{cos}}, \quad y = (I - P_{\text{LD}})r_{\text{cos}}. \quad (13)$$

The amplitude is the nonnegative ridge projection

$$\hat{\kappa}_s = \frac{\langle x_s, y \rangle_+}{\|x_s\|_2^2 + 1/N_{\text{cal}}}. \quad (14)$$

The projection is target-aware only through the schedule-derived λ_s . It never uses target loss values. The nonnegative truncation encodes the observed sign: the residual we want to transfer is a positive loss excess after LR drops.

3.5 Training and Testing Procedure

The calibration procedure is:

1. Load the source `cosine_72000.csv` curve for a source scale.
2. Compute the frozen MPL prediction and source residual r_{cos} .
3. For each target schedule, compute d_t , q_s , λ_s , and a_s from the LR schedule only.
4. Build the source response feature $\phi_{\lambda_s, \text{cos}}$, project it and the source residual away from MPL-LD tangent directions, and compute $\hat{\kappa}_s$.

The prediction procedure is:

1. Build the target response feature $\phi_{\lambda_s, s}$ from the target LR schedule.
2. Output Eq. (1).
3. Use the target loss curve only to compute evaluation MAE.

4 Why the Formula is Plausible

Linear response motivation. The response feature follows from a first-order tracking-error approximation. Let an unobserved excess-loss state e_t track an LR-dependent equilibrium. Around the current trajectory, a positive LR decrement perturbs the state by a forcing proportional to d_t , and the perturbation decays under the local optimizer dynamics:

$$e_t \approx \rho_t e_{t-1} + b d_t, \quad \rho_t \approx \exp(-\lambda \eta_t). \quad (15)$$

Projecting this state onto loss gives Eq. (5) up to a scalar amplitude. This derivation is only a local approximation, but it explains why the feature is causal, why constant LR creates no new response, and why the memory should be measured in cumulative LR rather than raw step count.

Why not fit the response directly? Directly fitting κ from cosine residuals assumes $r_{\text{cos}} \approx \kappa \phi_{\lambda, \text{cos}}$. The audit rejects this assumption: the no-nuisance raw projection worsens WSD-family MAE by hundreds of percent. The reason is not that the response feature is useless, but that cosine residuals contain a smooth component aligned with MPL’s LR-dependent parameter errors. Eq. (13) makes the identification statement explicit:

$$r_{\text{cos}} = \kappa \phi_{\lambda, \text{cos}} + J_{\text{LD}} \Delta \theta_{\text{LD}} + \epsilon. \quad (16)$$

After applying $I - P_{\text{LD}}$, the nuisance term is removed to first order, leaving a one-dimensional estimate of the transferable response amplitude.

Table 1: Main evaluation of the q_2 half-life MPL-LD response. Values are percentage MAE changes relative to frozen MPL; negative is better.

split	group	mean	median	worst	wins / non-harm
same scale	WSD-family	-29.88	-35.97	-4.67	15/15, 15/15
cross scale	WSD-family	-24.95	-23.21	-3.15	30/30, 30/30
all source-target	WSD-family	-26.59	-24.33	-3.15	45/45, 45/45
same scale	controls	+0.00	+0.00	+0.00	0/9, 9/9
all source-target	controls	+0.00	+0.00	+0.00	0/27, 27/27

Table 2: Same-scale WSD-family results by target.

target	mean	worst row	wins
WSD sharp	-46.27	-37.12	3/3
WSD linear	-39.73	-29.81	3/3
WSD-con 3e-5	-41.12	-35.97	3/3
WSD-con 9e-5	-13.40	-9.17	3/3
WSD-con 18e-5	-8.88	-4.67	3/3

Why q_2 , not a family label? The model needs to distinguish a single-step drop from a smooth distributed decay. A family label would solve this numerically but would not generalize. The concentration $q_s = \sum_t p_t^2$ is a schedule statistic with a direct meaning: it is the inverse effective number of drop atoms. This is more interpretable than the older max-drop ratio $q_\infty = \max_t p_t$, while producing nearly identical results in the current audit.

Why support projection, not a gate? The locality factor is often tempting to describe as a gate, but that is misleading. It is the exact energy fraction left when a uniform forcing on the LR-drop support is projected away from the uniform full-horizon mode. This matters because full-horizon cosine decay can have many small positive drops, but it should not receive a local cooldown transient correction.

5 Experiments

5.1 Main Results

Table 1 summarizes the current formula. Same-scale and cross-scale WSD-family transfer both improve every row. The controls remain unchanged because constant schedules have $d_t = 0$, and the short-cosine control has zero support-projection locality.

Table 2 shows the same-scale WSD rows by target family. The largest gains are on WSD sharp, WSD linear, and the lowest WSD-con endpoint; the high-end WSD-con targets still improve, but the effect is smaller because the residual there is weaker.

5.2 Ablations

Table 3 gives the main ablations. The no-nuisance raw projection is the most important negative control: it fails catastrophically, confirming that cosine residual contamination is a real identification problem. Removing locality gives slightly stronger WSD means but damages controls, so it is not acceptable. A density-projection locality rule is mathematically natural, but it keeps too much of the smooth cosine decay and also harms controls. The older q_∞ response rule is numerically almost identical to q_2 , but q_2 has the cleaner effective-drop-count interpretation.

5.3 Schedule Diagnostics

The schedule-derived quantities in Table 4 show that the formula behaves as intended. WSD sharp and WSD linear have long but finite drop supports, so $a_s \approx 0.817$ and $\lambda_s \approx 9.03$. WSD-con curves have a single-step

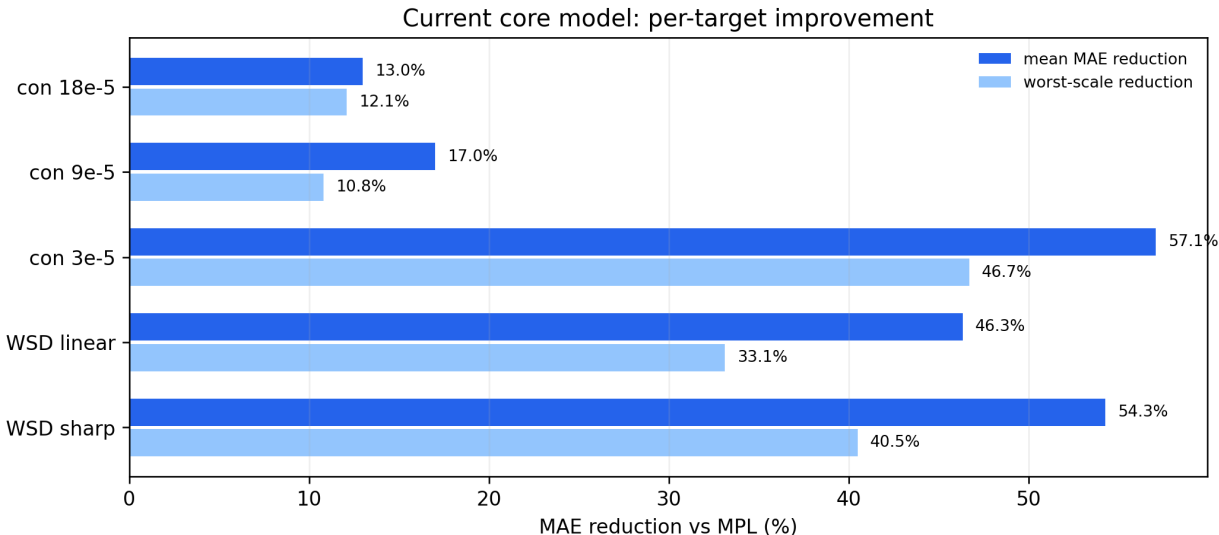


Figure 1: Per-target MAE changes for the current formula and references. The current mechanism-native estimator improves every WSD-family row; high-end WSD-con rows have smaller but still positive gains.

Table 3: Theory-refinement and safety ablations.

variant	WSD mean	WSD worst	WSD wins	control reading
q_2 half-life + support projection	-29.88	-4.67	15/15	9/9 non-harm
old q_∞ + support projection	-29.87	-4.67	15/15	9/9 non-harm
q_2 density projection	-30.22	-4.67	15/15	worst +8.25, 6/9 non-harm
q_2 no locality	-30.88	-4.67	15/15	worst +56.99, 6/9 non-harm
no-nuisance raw projection	+602.17	+2366.35	0/15	fails on WSD
old MPL-LD fixed ridge	-27.25	-3.00	15/15	9/9 non-harm
DCT performance reference	-32.83	-5.30	15/15	cross-scale worst +26.68

drop, so $q_s = 1$, $a_s \approx 1$, and $\lambda_s = 18.05$. The short-cosine control has a diffuse full-horizon support and is projected to zero by the locality factor.

5.4 Residual Curves

The residual plots in Figure 4 compare MPL, the current MPL-LD response, the previous fixed-ridge MPL-LD reference, and a DCT performance reference. The useful pattern is not that every point is fitted perfectly; rather, the current response systematically reduces WSD-family residuals while leaving safety controls unchanged.

6 Discussion

What the current result supports. The strongest supported claim is not that we have discovered a universal optimizer law. The supported claim is more specific: after removing local MPL-LD tangent drift from the cosine residual, a one-dimensional causal LR-drop response transfers reliably from cosine to the current WSD-family targets. The response amplitude is low capacity, the response shape is schedule-only, and the main evaluation uses no target-loss fitting.

Why the model is not just overfitting. The formula has only one residual-fitted scalar for a source scale and response operator. The response rate, locality factor, and nuisance subspace are computed from the LR schedule and the frozen MPL formula. Several higher-capacity alternatives were intentionally rejected:

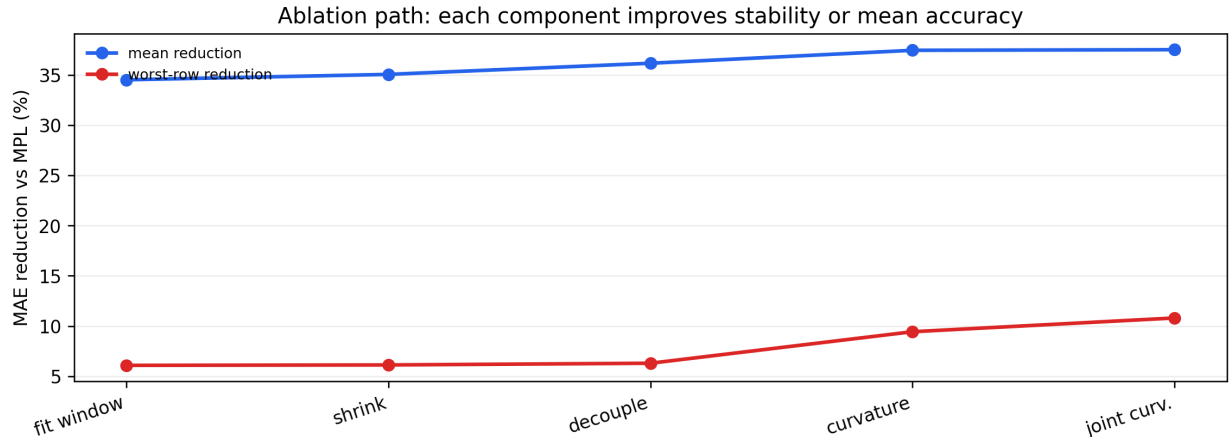


Figure 2: Ablation summary. DCT remains a useful numerical reference, but the MPL-LD tangent projection is the main interpretable candidate because its nuisance directions are tied to MPL parameter error and its cross-scale audit stays non-harmful.

Table 4: Schedule-only diagnostics used by the current formula.

curve	q_2	n_{eff}	support span	a_s
WSD sharp	3.52e−4	2842.3	4000	0.8168
WSD linear	2.50e−4	3999.0	4000	0.8168
WSD-con targets	1.0000	1.0	2	0.9999
Cosine 24k control	5.65e−5	17702.8	21840	0.0000
Constant controls	0.0000	0.0	0	0.0000

DCT projection is numerically strong but less interpretable and less stable cross-scale; density projection is natural but harms controls; no-locality improves WSD means but damages the short-cosine control; raw projection fails completely. These negative results are part of the model selection evidence.

Remaining limitations. The current result is still a development candidate. The q_2 half-life bracket uses the logging interval as a structural prior; it is cleaner than the older endpoint constants, but it is not a theorem about training dynamics. The suffix start 8000 is a protocol choice that removes early cosine transients; later starts remain positive but reduce the gain. The support-projection locality explains the safety boundary for current controls, but it is not a complete dynamical derivation of all schedule families. Finally, all main numbers come from the current loss-curve repository; external schedules and new training runs are needed before making broader claims.

7 Conclusion

This paper rewrites the current project around one core line: MPL is the baseline, cosine residuals are contaminated by MPL LR-dependent drift, and a causal LR-drop response becomes useful only after MPL-LD tangent nuisance projection. The final working formula is

$$\hat{L}_s(t) = L_{\text{MPL},s}(t) + a_s \hat{\kappa}_s \phi_{\lambda_s,s}(t), \quad \lambda_s = \frac{\lambda_{\text{obs}}}{2 - q_s}, \quad q_s = \sum_t \left(\frac{d_t}{\sum_u d_u} \right)^2.$$

With support-projection locality, the method improves all current WSD-family rows (−29.88% same-scale mean, −24.95% cross-scale mean) while leaving controls non-harmful. The main value of the result is interpretability under a strict cosine-only calibration protocol: it keeps the correction small, schedule-derived, and explicit about what has not yet been externally validated.

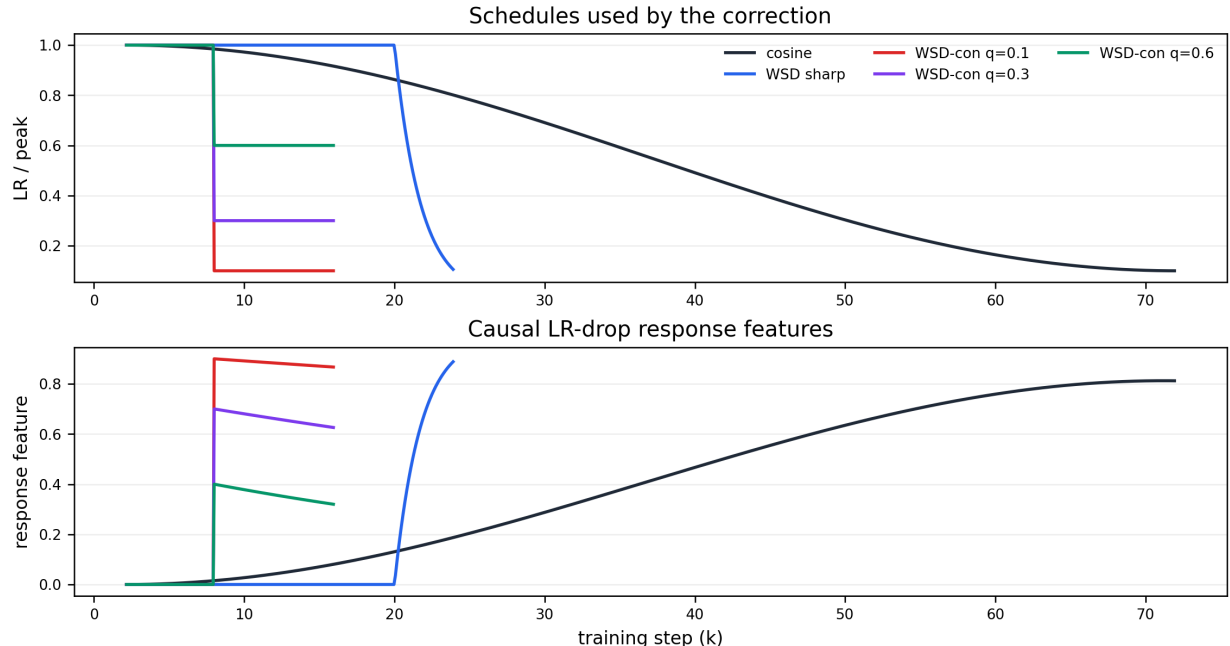


Figure 3: Schedule features used by the formula. The construction separates concentrated LR-drop responses from diffuse full-horizon decay without using target loss.

A Implementation Details

Finite-difference tangent basis. The MPL-LD tangent basis is computed by perturbing the four LR-dependent MPL directions

$$\log B, \quad \log C, \quad \log \beta, \quad \log \gamma$$

around the frozen MPL parameters, then evaluating the resulting loss-curve differences on the source cosine suffix. The projection is an ordinary least-squares orthogonal projection onto these four columns.

Cross-scale evaluation. For same-scale rows, the source and target scale are identical. For cross-scale rows, a source cosine residual from one scale is used to calibrate $\hat{\kappa}_s$, and the result is applied to a target curve at another scale. There are $3 \times 2 \times 5 = 30$ cross-scale WSD rows. The current formula improves all of them.

Reproduction artifacts. The main artifacts used in this paper are:

- results/interpretable_theory_refinement/REPORT.md
- results/interpretable_error_model/
- paper/figs/

The first path contains the current q_2 half-life audit; the second keeps historical fixed-candidate and error-curve audits; the third contains copied figures for a self-contained paper build.

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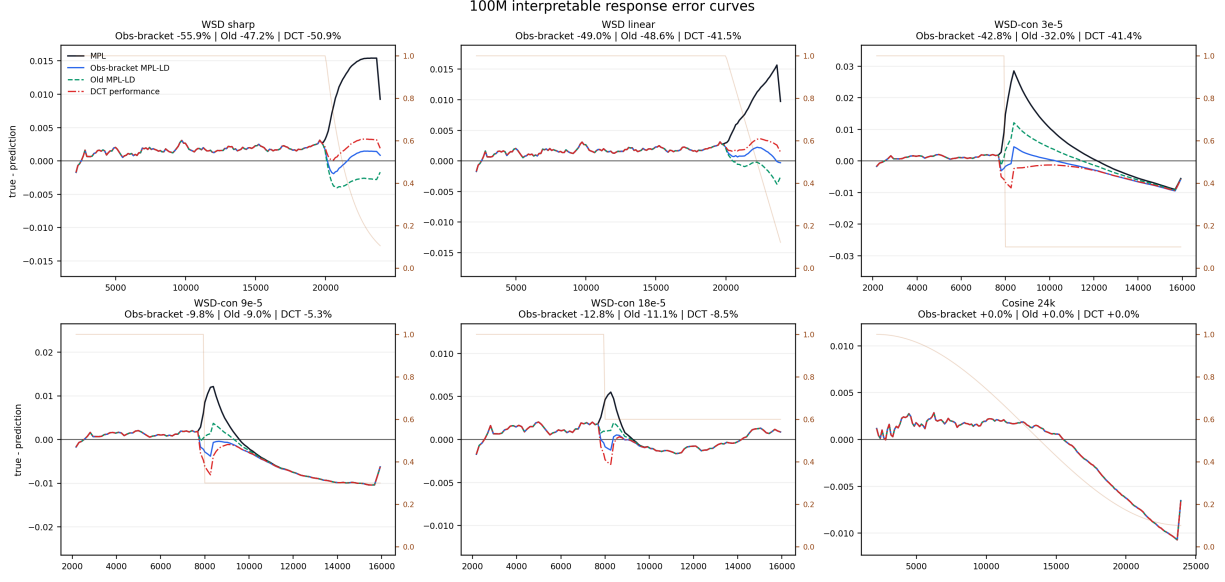


Figure 4: Representative error curves at 100M. The correction is fitted from the source cosine residual only and evaluated on WSD-family targets and controls.

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